MILITARY SPECIFICATION

BORON FILAMENT REINFORCEMENT, CONTINUOUS, EPOXY RESIN IMPREGNATED

1. SCOPE

1.1 Scope. This specification covers continuous 4-mil diameter boron filament-reinforced epoxy resin matrix prepreg of two types. All material to this specification is of one class (non-integral, woven glass fabric carrier) for making low pressure laminates for use up to 375°F.

1.2 Classification. The resin impregnated reinforcement shall be of the types, as specified:

Type I - General purpose (for use at temperatures to 160°F).
Type II - Heat resistant (for use at temperatures to 375°F).

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-M-261	Methyl Ethyl Ketone

Military

MIL- R-9300	Resin, Epoxy, Low Pressure Laminating
MIL-B-83353(USAF)	Boron Monofilament, Continuous, Vapor Deposited

STANDARDS

Federal

FED. Test Method STD No. 406	Plastics; Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by supplies in connection with specific procurement functions

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should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents of the issue shown form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D 579 Standard Specifications and Methods of Test for Woven Glass Fabrics

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

SOCIETY FOR PLASTICS INDUSTRY (SPI)

Prepreg 1 Resin and Volatile Content of Pre-impregnated Inorganic Reinforcements
Prepreg 2 Flow of Pre-impregnated Products

(Copies of Society for Plastic Industry publications may be obtained from the Society of the Plastics Industry, Incorporated, 250 Park Avenue, New York, N.Y. 10017.)

3. REQUIREMENTS

3.1 General. The materials covered by this specification shall consist of boron filaments collimated parallel and in-plane, preimpregnated with a thermosetting epoxy resin and bonded to a woven glass fabric carrier with the same resin. They shall be capable of being molded, using low pressure laminating methods (under 200 psi), to produce a cured molding having properties described in this specification.

3.1.1 Yield. The minimum number of square feet of material per pound of filament supplied, shall be based on the number of filaments per inch of width as stated below. A maximum loss of 7 percent shall be allowed for processing.

Filaments per inch of tape width	Square feet of tape per pound of bare filament
206 208 210 212 214	26 ± 0.5

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3.2 Materials. The component materials shall be as specified in 3.2.1, 3.2.2 and 3.2.3 and shall be of high quality, entirely suited for the purpose.

3.2.1 Impregnating resin. The resins for types I and II materials shall conform to the general requirements of MIL-R-9300 including the additional properties shown in table I of this specification. The resin shall be free of foreign materials, noncorrosive to metals and shall be capable of being molded, using low pressure laminating methods, to a fully thermoset state and meet the requirements of this specification.

3.2.2 Boron filament. The boron filament used shall meet the requirements of MIL-B-83353(USAF).

3.2.3 Glass fabric carrier. Unless otherwise specified, the glass fabric carrier shall conform to the requirements of ASTM D579, type 58 (industry style 104) fabric with A-1100 soft epoxy-compatible finish. These supporting materials shall be capable of holding the collimated boron filaments in their fixed positions while in storage, during shipping, and when handling during lay-up and fabrication operations. The warp of the carrier cloth shall be positioned parallel to the length of boron filament in the prepreg.

3.2.4 Reinforced, uncured, impregnated material. Boron-filament-reinforced, uncured, impregnated material shall exhibit the following characteristics:

a. Boron filaments shall be fully wetted by the resin.
b. There shall be 210 $\pm$ 4 filaments per 1.000 inch width of material, and the filaments shall be spaced at 0.0002 inch, minimum, and 0.012 inch, maximum, intervals, with no filaments in contact or crossing over others.
c. Variation from the spacing requirements of (b), above, shall be limited to 3 percent, maximum of the lot of material inspected, except that any lot of material exhibiting a spacing gap in excess of 0.040 inch shall be adjudged unacceptable.
d. All boron filaments shall be collimated and parallel to the centerline of the matrix within an angle of 15 minutes.
e. The boron filaments shall not be crimped.
f. Except for resin used for splicing filaments, there shall be no cured resin particles in the material.
g. The impregnated material shall be free of parting agents and other foreign matter.
h. The physical properties of the uncured material shall be in accordance with table I.

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3.2.4.1 Physical properties. The as-received, uncured preimpregnated material shall meet the requirements of table I.

Table I. Physical properties (types I and II)

Property	Requirement
Tack	shall adhere to a vertical mold surface at a temperature within the range 68 to 78°F for at least 8 hours
Flow	±3 percent of that specified
Volatile content	range from 0.3 percent minimum to 2 percent maximum
Resin content	±2 percent by weight of a value specified by the purchaser within a range of 25 to 45 percent.

3.2.4.2 Splices. Boron filament splices in the prepregged material shall be kept minimal to assure adherence to and attainment of the mechanical properties shown in table III of this specification.

3.2.4.3 Impregnated tape width. Impregnated material shall be furnished in the width specified. The width shall not vary more than ±.005 inch per inch of ordered width.

3.2.4.4 Impregnated tape length. Impregnated length per roll shall be 125-500 feet. The last roll of a batch shall exceed 25 feet.

3.2.4.5 Impregnated tape uniformity. Each batch of impregnated material shall be of uniform quality throughout. Any section of impregnated material which does not meet the requirements of this specification shall not be removed from a continuous length of material. Linear footage of rejected material shall be itemized on the inspection tag and attached to the roll of material and shall not be included in the specified length of impregnated material. Maximum acceptable waviness of any 24 inch length shall be .030 inch from the edge.

3.2.4.6 Cured prepreg physical properties. After fabrication of a cured composite in accordance with 4.8, the laminate shall meet the requirements of table II.

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Table II. Physical properties for cured composites.

Property	Average test value 1/
Per ply thickness	0.0051 to 0.0054 inch
Specific gravity, minimum	1.94

1/ Based on three test samples.

3.2.5 Mechanical properties. After fabrication of a cured test laminate in accordance with 4.8, the laminate shall meet the requirements of table III.

3.3 Storage life. The prepreg material shall be capable of meeting the requirements of this specification after storage of 6 months at temperatures not exceeding 0°F, plus a minimum of 15 days storage at 75°F maximum.

3.4 Workmanship. Prepreg material furnished to this specification shall be of quality workmanship and shall be free of all impurities and defects which could adversely affect its performance. Visible indication of dry spots, voids, crossed or broken filaments, irregular carrier or incomplete impregnation shall be marked by inserts and shall be cuase for rejection. The total roll shall be rejected if such areas exceed 2 percent of the total tape.

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Table III. Mechanical properties.

Property/test temperature	Type I	Type 2
	Minumum average requirement.	
Longitudinal flex. str. psi		
75°F	240,000	240,000
160°F	200,000	
270°F		200,000
375°F		170,000
Longitudinal flex. mod. psi		
75°F	27.5×10^6	27.5×10^6
160°F	25.0×10^6	
270°F		26.0×10^6
375°F		20.0×10^6
Transverge flex. str. psi		
75°F	13,000	13,000
160°F	9,000	
270°F		10,000
375°F		8,000
Shear str. horz. psi		
75°F	13,000	13,000
160°F	9,000	
275°F		7,000
375°F		5,000
Longitudinal ten. str. psi		
75°F	205,000	205,000
Longitudinal ten. mod. psi		
75°F	30×10^6	30×10^6
Transverge ten. str. psi		
75°F	10,000	10,000
Transverge ten. strain μ inches		
75°F	4,000	4,000

NOTES: a. All property testing accomplished on unidirectional laminates.
b. Five individual specimens shall be tested for each property and each test temperature.
c. Specimens shall be taken from laminates with a boron filament content of 50 to 55 percent by volume.
d. Elevated temperature tests to be conducted after 30 minutes at the specified temperature.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the procuring agency. The procuring agency reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The classification of test and inspection shall be as specified:

a. Quality conformance test (4.10)
b. Acceptance test (4.11)

4.2.1 Certificate of conformance. Unless otherwise specified in the purchase order, the supplier shall furnish with each shipment, a certified report (in triplicate) signed by a duly authorized representative of the supplier, stating conformance to the requirements specified herein and listing the specific results of all the acceptance tests. This report shall also include this specification number, the purchase order number, and the batch number.

4.3 Subcontractor. When materials for subcontract fabrication are purchased directly by the subcontractor, the subcontractor shall be responsible for determining that the material meets all the requirements of this specification. With each part shipment the subcontractor shall submit a report as specified in 4.2.1.

4.4 Inspection records. The supplier's inspection records of examination and tests for conformance to the requirements of this specification shall be kept complete and available to the purchaser upon request.

4.5 Reinspection. The purchaser reserves the right to perform or witness any of the inspections specified herein, when these sinspections are deemed necessary to substantiate material conformance. Tested materials which do not meet these requirements shall be subject to rejection.

4.6 Inspection lot. An inspection lot may consist of material from more than one batch, forming part of one purchase order and submitted for acceptance at one time. Under these circumstances, each batch shall be

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subjected to the herein specified acceptance tests.

4.6.1 Batch. A batch shall be that quantity of material compounded and manufactured at one time, from one batch of resin, in one continuous manufacturing operation.

4.7 Test conditions.

4.7.1 Standard conditions. Normal temperature tests shall be conducted at 70° to 80°F (21° to 27°C). If the validity of property values determined within this temperature range is questionable, the tests shall be repeated at standard conditions (23° ± 1°C and 50 ± 5% relative humidity).

4.8 Test specimen preparation. Test panels from which test specimens will be prepared shall be fabricated in a manner defined by the prepreg materials supplier, consistent with standard aircraft processing procedures and compatible with the physical and mechanical requirements of this specification. Processing shall include post cure if required of the materials system selected to develop optimum physical and mechanical properties.

4.8.1 Completed test specimens. Test specimens shall be cut from the test panels prepared as in 4.8 using a 120 grit or finer diamond wheel operated at 4000, minimum, sfpm and 0.5 to 1.0 inch per minute feed rate or using other suitable cutters and machining conditions which will provide finishes equivalent to use of diamond wheels.

4.9 Testing.

4.9.1 Uncured prepreg physical test specimens.

Test	Specimen number and size
Volatile content	1 ply, 3 inch square
Resin content	1 ply, 3 inch square
Flow	4 plies, 3 inch square stacked 0 degrees and 90 degrees alternately
Tack	2 plies, 4 inch long times width and thickness of prepreg

4.9.2 Cured prepreg mechanical test specimens. Unless otherwise specified herein, all filament orientation within the length of the specimen shall be within ±½ degrees. Specimens shall be 15 plies thick. A mean value for strength, based on three specimens for acceptance shall be reported.

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4.9.2 Cured prepreg mechanical test specimens (continued)

Test	Paragraph	Filament direction
Flexure, longitudinal	4.10.2.1	0°
Flexure modulus	4.10.2.1	0°
Flexure, transverse	4.10.2.2	90°
Shear, horizontal	4.10.2.3	0°
Tension, longitudinal	4.10.2.4	0°
Tension modulus	4.10.2.4	0°
Tension transverse	4.10.2.5	90°

4.10 Quality conformance test

4.10.1 Physical tests, prepreg material. Physical tests shall be performed on the prepreg material using specimen sizes and quantity required for test as specified in 4.9 and tested as described below.

4.10.1.1 Resin content and flow. Perform tests in accordance with procedures described in Society of Plastics Industry methods of test for prepregs, Standards Numbers SPI-Prepreg 1 and 2. Determine resin content by dissolving in boiling MEK for six minute intervals until there is no further loss of weight (see TT-M-261).

4.10.1.2 Volatile content. The specimen shall be weighed to the nearest 0.01 gram and placed in an air circulating oven at 325°F ± 5°F for 15 ± 1 minute, then removed and cooled in a desicator to standard conditions and reweighed to the nearest 0.01 gram. The volatile conetnt shall be calculated as follows:

$$\text{Percent Volatile Content} = \frac{W_1 - W_2}{W_1} \times 100$$

Where W_1 = initial weight

Where W_2 = weight after exposure

The mean percent volatile content shall be based on the test results of three specimens.

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4.10.1.3 Tack. With the packaging separator sheet up, place a 3" x 4" prepreg on a 0.125 x 4 x 8 stainless steel plate, held vertically, having a smooth (RMS 100 maximum) surface and maintained at standard conditions. Remove air bubbles and wrinkles with a squeegee. Remove the separator sheet. Prepreg shall not be dislocated by its own weight in less than eight hours, but shall be capable of removal without loss of more than five percent of the initial weight.

4.10.1.4 Boron filament count. Perform actual filament count, using an optical comparator at 50-100X according to the requirements of 3.2.4.

4.10.1.5 Boron filament spacing. Perform test using an optical comparator at 50-100X according to the requirements of 3.2.4. for determining maximum spacing between filaments.

4.10.2 Mechanical tests, cured composite laminates. Mechanical tests shall be performed on cured composite laminates made from material supplied to this specification and cured as recommended by the manufacturer. Specimen sizes and quantity required for test are described in 4.9.

4.10.2.1 Flexure, longitudinal (0°) strength and modulus.

1. Specimen dimensions shall be as shown in figure 1. The thickness dimensions (t) will be in the range 0.0765 to 0.0810 inch. A variation in thickness over a specimen may not exceed 0.004 inch.

2. A load support method shall be utilized as shown in figure 1.

3. The specimen shall be loaded in a universal test machine at a load rate of 0.05 inch per minute.

4. During the test, the deflection of the specimen midpoint is to be graphically recorded as a function of the applied load. The deflectometer is removed prior to specimen failure.

5. Record the load at failure.

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6. Calculations:

A. Flexure strength, ultimate - $f_u = \frac{3\ PS}{2\ Wt^2}$

P = load
S = span
W = specimen width
t = specimen thickness

B. Flexure modulus, $E_f = \frac{S^3}{4\ Wt^3} \times \frac{p}{d}$

Where p = applied load
d = midpoint deflection at load p

$\frac{p}{d}$ = initial slope of the load-deflection curve

$\frac{s}{W}$ = span between reaction points
W = specimen width

t = specimen thickness

4.10.2.2 Flexure, transverse (90°)

1. Specimen dimensions shall be as shown in figure 2. The thickness dimension (t) may vary from .0765 to .0810.

2. A load support method shall be utilized as shown in figure 2.

3. The specimen shall be loaded on a universal test machine at a load rate of .05 inch per minute.

4. Record the load at failure.

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5. Calculations:

Flexure, ultimate = $f_u = \frac{3\ PS}{4\ Wt^2}$

P = load
S = span
W = specimen width
t = specimen thickness

4.10.2.3 Horizontal shear

1. Specimen dimensions shall be as shown in figure 3. Thickness may vary from .0765 to .0810. Variation in a specimen shall not exceed 0.003 inch.

2. The specimen shall be loaded to failure in a universal test machine at a load rate of 0.05 inch per minute.

3. A load support method shall be utilized as shown in figure 3.

4. Record the load at failure.

5. Calculations:

Horizontal shear - $F_{Hs} = \frac{3\ P}{4\ Wt}$

P = load
W = specimen width
t = specimen thickness

4.10.2.4 Tension, longitudinal (0°) strength and modulus

1. Specimen dimensions shall be as shown in figure 4. The thickness dimension (t) will be in the range of .0294 to .0330 (6 plies).

2. Specimens may be individually molded or cut (diamond tool recommended) to width required.

3. 1/16 inch bonded Scotch ply $0^\circ/90^\circ$ "S" glass tabs shall be bonded on both ends as shown in figure 4. Adhesives used shall be appropriate to withstand the temperature environment of the test.

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4. Self-aligning grips shall be used, completely covering the tab area.

5. The tensile specimen shall be instrumented with an appropriate strain gage to obtain the load-deflection curve to obtain geographically the ratio of the stress to strain required to calculate the tensile modulus.

6. Record the load at failure.

7. Calculation

A. Tensile strength, ultimate $F_t = \frac{P}{A}$

F_t = Ultimate tensile strength, psi
P^t = load, lbs
A = Specimen cross-sectional area, inches

B. Modulus of Elasticity, tensile $E = \frac{F_t}{\epsilon}$

E = Modulus of elasticity in tension, psi
F_t = Tensile stress, psi
ϵ = Corresponding strain in inches per inch

4.10.2.5 Tension strength, transverse

1. This test is similar in all respects to the longitudinal tensile test described in paragraph 4.10.2.4 and shown in figure 4 with the following exceptions

a. The orientation of the boron filament shall be at 90° to the long dimension of the specimen.

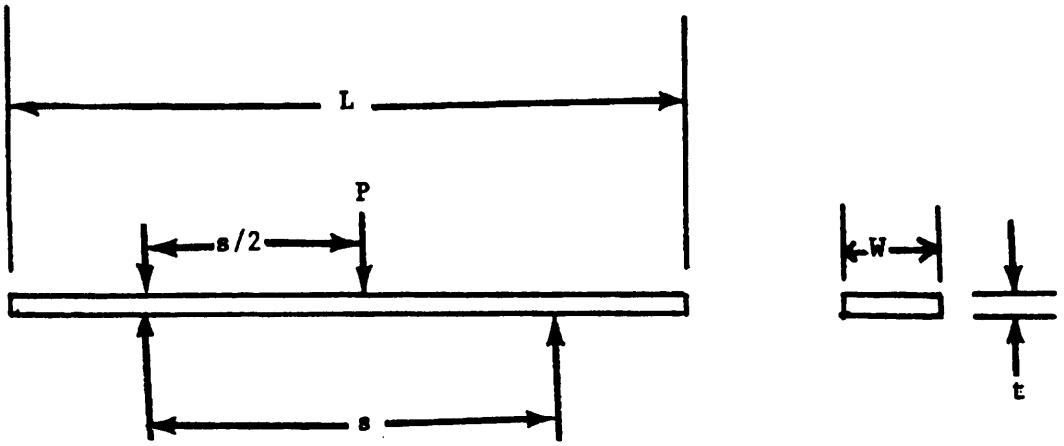
b. Failure will occur at significantly lower load values.

c. All other aspects may remain the same and the calculations for the strength are also the same.

Tensile strength, transverse $F_t = \frac{P}{A}$

F_t = ultimate tensile strength transverse, psi
P^t = load lbs
A = cross-sectional area, square inches

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Specimen Dimensions

Length (L) = 4.0
Width (W) = 0.500
Thickness (t) = 0.0765 to 0.0810
Span (s) = 2.92

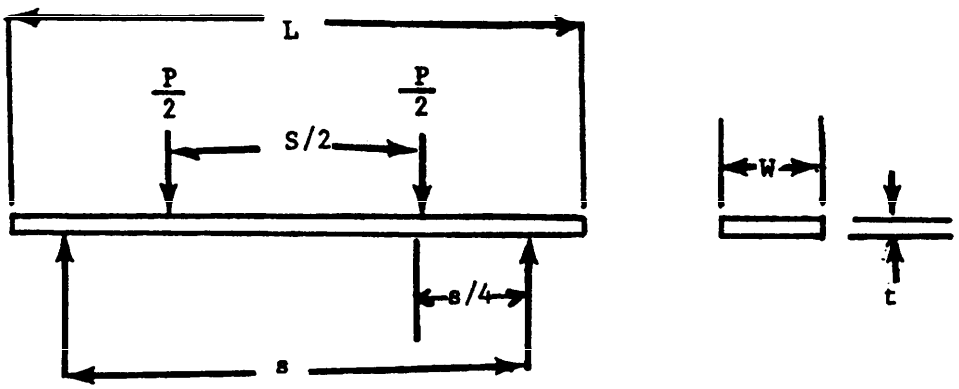
All filaments shall be 0° to the L dimension.

Load and reaction supports shall be 1/8" radius steel rod with length=W min.

Dimensions are in inches.

Figure 1
Longitudinal flexure (0°) .

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Specimen Dimensions

Length (L) = 3.0 ± 0.1
Width (W) = 0.500 ± 0.020
Thickness (t) = 0.0765 to 0.0810
Span (s) = 2.00

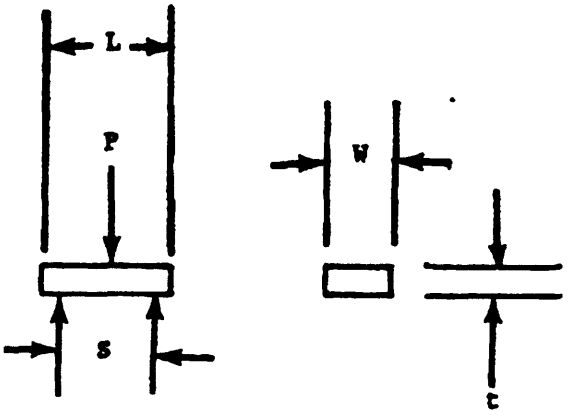
Load and reaction supports shall be 1/8" radius steel rod with length=W min.

All filaments shall be 90° to the L dimension.

Dimensions are in inches.

Figure 2
Transverse flexure (0°)

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Specimen Dimensions

Length (L) = 0.60 ± 0.01
Width (W) = 0.250 ± 0.003
Thickness (t) = 0.0765 - 0.0810
Span (s) = 4t

Load and reaction supports shall be 1/8" radius steel rod with length = W min.

All filaments to be 0° to the L dimension.

Dimensions are in inches.

Figure 3

Horizontal shear (0°)

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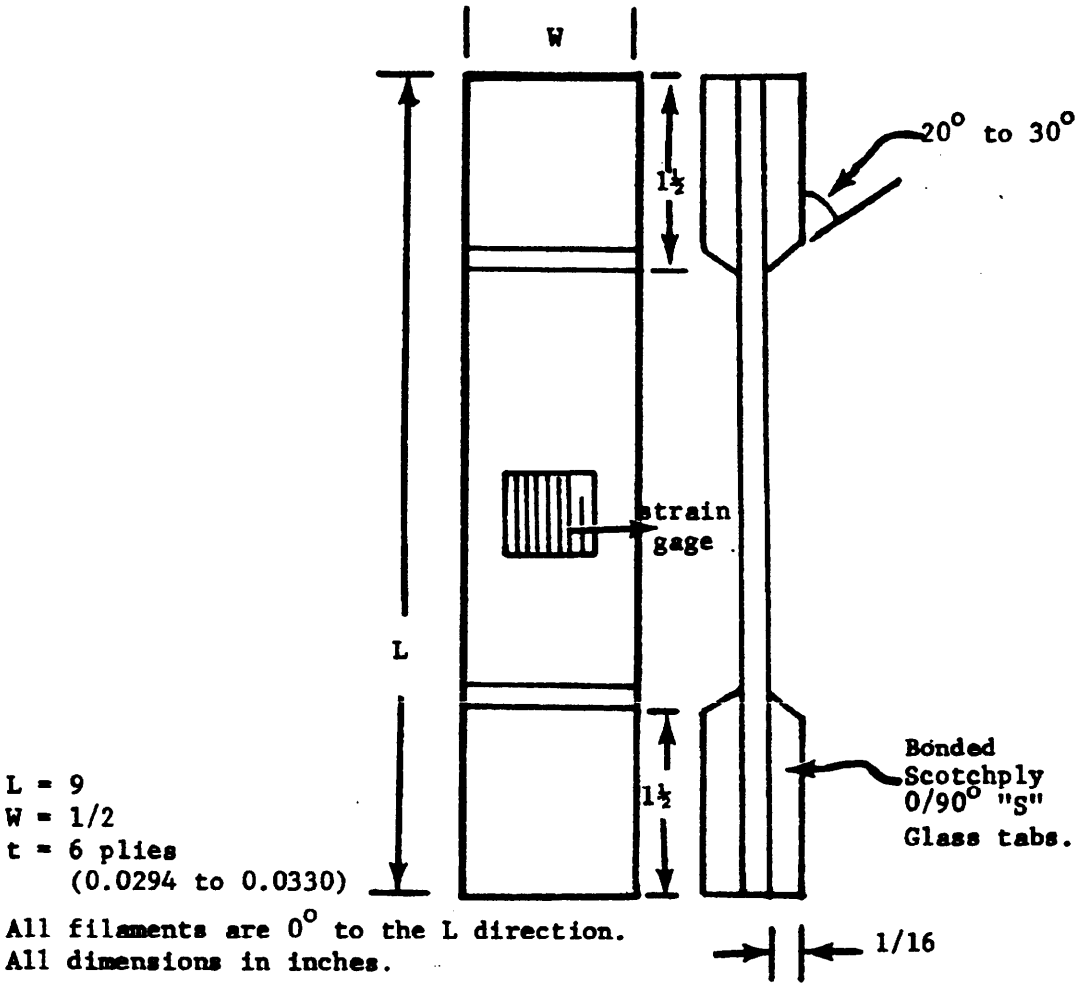


Figure 4
Longitudinal tensile specimen.

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4.10.2.6 Optional tensile test 0° and 90° composites.

1. Preparation of test specimen.

a. Prepare boron composite face sheets 1.0 inches wide by 22 inches long with thickness as follows:

(1) All fibers 0° to 22 inches dimension - 6 plys (0.0306 to 0.0324)
(2) All fibers 90° to 22 inches dimension - 8 plys (0.0408 to 0.0432)

b. Bond boron composite face sheet into sandwich beams as follows:

(1) For 0° composites use
(a) 23 $lb./ft.^3$ aluminum honeycomb core 1.5 inches thick by 1.1 inches wide by 22 inches long
(b) 0.125 inch thick aluminum sheet same width and length as the boron skin for the opposite face

(2) For 90° composites use
(a) 4.5 $lb./ft.^3$ aluminum honeycomb core 1.5 inches thick by 1.1 inches wide by 22 inches long
(b) 0.080 inch thick epoxy glass fabric laminate same width and length as the boron skin for the opposite face

(3) Bond using two layers of adhesive and cure.

2. The load support method utilized shall be as shown in Figure 5.

3. The test procedure shall be as follows:

a. Load the specimen to failure at the following load rates:

(1) 0° - 700 pounds per minute.
(2) 90° - 70 pounds per minute.

b. Record the load at failure.

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4. The stress in the composite facing is :

$$\sigma_t = \frac{4P}{Wt\left[C + \frac{(t + T)}{2}\right]}$$

where σ_t = stress in psi
P = load in pounds
W = width of composite facing in inches
t = thickness of composite facing in inches
C = thickness of core in inches
T = thickness of opposite facing in inches

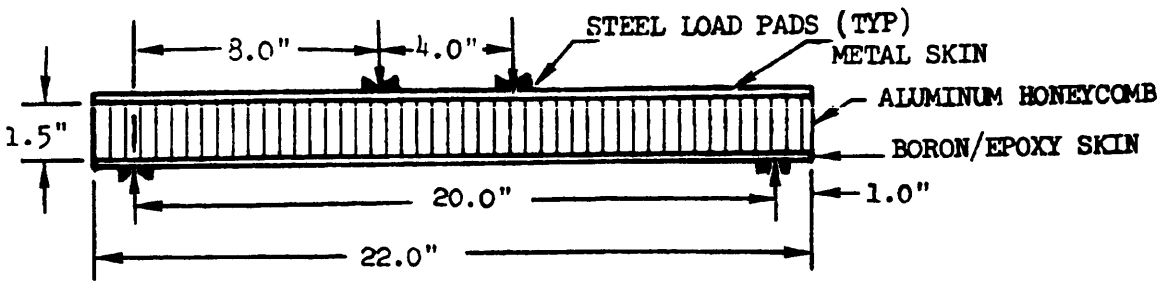


Figure 5
Longitudinal (0°) and Transverse (90°) Sandwich Beam Tensile Specimen

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4.11 Acceptance tests. The following shall be classified as acceptance tests. A random sample from each batch of production material shall be selected for inspection. Inspections shall be made for conformance to requirements for identification marking (5.1.1), and packaging and packing (5.1 and 5.2). The roll of prepreg shall then be allowed to reach standard temperature before breaking the seal. Open seal and remove 5 square feet of prepreg. Reseal the package of prepreg and return to 0°F storage. Identify the 5 square foot sample and proceed to inspect for the following requirements:

(1) boron filament count, spacing, and uncured prepreg physical properties (3.2.4)
(2) boron filament splices (3.2.4.2)
(3) prepreg width (3.2.4.3)
(4) prepreg length (3.2.4.4)
(5) uniformity (3.2.4.5)
(6) workmanship (3.4)
(7) cured prepreg strength at 0° and 90° to filament direction - see table III at R.T. tensile strength and strain , and 0° and 90° tensile strength at 160°F (type I).

4.12 Retest. If a sample fails to meet two or less requirements of this specification, a retest shall be permitted. A second failure shall be cause for rejection. The results of both the original tests and the retests shall be included in the test report.

4.13 Rejection. Each batch of material shall be rejected if it does not pass the acceptance tests subject to the retest provisions of 4.12.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Prepreg material shall be rolled on a reel of not less than 8 inches in diameter. A nonadherent, nonporous, and nontransferable paper or separator of a contrasting color, shall be used on one side of the material against the glass carrier to prevent the layers of material from sticking to each other. Each roll or rolls of prepreg shall be heat sealed in an evacuated, moisture-proof plastic bag. An identification tag shall be placed within each bag prior to sealing. (See 5.2)

5.1.1 Identification marking. Each roll of prepreg shall be permanently marked with the following data prior to packing:

a. SPECIAL INSTRUCTIONS (as required, see 5.3)

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b. RESIN TREATED BORON REINFORCED PREPREG. LOW PRESSURE MOLDING

c. PURCHASER DESIGNATION NO.

d. DATE OF IMPREGNATION

e. RESIN CONTENT, PERCENT

f. VOLATILES, PERCENT

g. FLOW, PERCENT

h. WIDTH OF MATERIAL

i. LINEAR FEET

j. MANUFACTURER'S BATCH NO. AND DATE OF MANUFACTURE

k. MANUFACTURER'S DESIGNATION

l. STORAGE TEMPERATURE, MAX: $0^{\circ}F$

m. SHELF LIFE: 6 MONTHS

n. INSPECTION RECORD AND COMMENTS: ITEMIZED DESCRIPTION OF REJECTED MATERIAL INCLUDING LINEAR FOOTAGE OF SUCH REJECTS AND THEIR LOCATION.

5.2 Packing. Units packaged as specified in 5.1 shall be packed in exterior-type shipping containers in a manner that will allow solid carbon dioxide to be packed in sufficient quantities to maintain a material temperature of $0^{\circ}F$, minimum, during transit. Upon receipt, containers shall be opened and examined to ascertain that solid carbon dioxide remains therein, if used. Prepreg rolls shall be packed in a horizontal position and containers so marked so as to ensure horizontal positioning for shipment and later stored in an upright, vertical position. The shipping container shall be so constructed as to assure safe delivery and acceptance at their destination. Shipping containers shall comply with carrier regulations applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified, marking of shipping containers shall be in accordance with MIL-STD-129 and include the following additional information.

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a. MIL-B-83369(USAF)
b. PURCHASE ORDER NO.
c. MANUFACTURER'S NAME TRADEMARK OR SYMBOL
d. MANUFACTURER's PART NO.
e. MANUFACTURER's BATCH AND LOT NO.
f. NOMINAL WIDTH AND LENGTH PER ROLL
g. STORAGE TEMPERATURE MAXIMUM 0°F
h. DATE OF MANUFACTURE
i. SHELF LIFE: 6 MONTHS
j. CAUTION: F R A G I L E

6. NOTES

6.1 Intended use. The material procured in accordance with this specification is intended for use in airframe and aerospace primary structural components and in similar assemblies where high stiffness and strength-to-weight ratios are required.

6.2 Procurement. Preimpregnated boron material shall be supplied by the linear foot in the width ordered.

Custodian:
Air Force - 11

Reviewers:
Air Force - 84

Preparing activity:
Air Force - 11

Project No. 9330-F456

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OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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